Final Report (FSP)

Project Title: Visualizing Data Distributions and Analyzing Relationships

1. Student Information

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- **Date of Submission: Sunday, 12th October 2025**
- **Course: Foundation of Statistics and Probability**

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2. Project Overview

- **Objective of the Project:**
  - The objective of the project is to explore given datasets and create Visualization for Boxplots, Correlation Coefficients, statistical significance, etc and generate meaningful insights from analysis.
- **Dataset Selected:**
  - Video Game Review Data
- **Tools Used:**
  - I used **Google Sheets** for Statistical calculations like min, max, mean, median,etc. Used **Orange data mining** tool for box plots and Correlation coefficients also used **Google Docs** for final report of my findings.

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3. Exploratory Data Analysis (EDA)

3.1 Dataset Description

- Numbers of records are **60** and attributes are **3** (Player ID, Hours Played, Review Score) in the provided Video Game Reviews Dataset.
- Two types of variables Present in the Video Game Reviews Dataset:
  1. **Categorical (Player ID)**; Unique ID
  2. **Continuous (Hours Played, Review Score)**; Quantitative data (number of hours) and Quantitative rating (0–100 scale).

3.2 Summary Statistics

- **Hours Played attribute:**

1. **Mean** of hours played attribute is **16.33** which shows average of hours played is 16.33.

2. **Median** is **16** which shows the middle value of hours played is 16.
3. **Mode** is also **16** which shows the most frequent value is 16.
4. **Min** is **5.**
5. **Max** is **45** which shows outlier value by which Mean is attracted above.
6. **Standard Deviation** is **7.71**.

- **Review Score attribute:**

1. **Mean** of Review Score attribute is **80.615** which shows average of review score is 80.615.
2. **Median** is **80.5** which is the middle value of the review score.
3. **Mode** is **89** which shows the most frequent value in the data set review score is 89.
4. **Min** is **63.3** which is the minimum review score.
5. **Max** is **100** which is the highest review score in the data set.
6. **Standard Deviation** is **9.913**.

Tables for Summary Statistics:

Hours Played	
Mean	16.33333333
Medain	16
Mode	16
Min	5
Max	45
SD	7.710144466

Review Score	
Mean	80.615
Medain	80.5
Mode	89
Min	63.3
Max	100
SD	9.913931731

3.3 Five-Number Summary

Review Score:

- Minimum Value of review score attribute is **63.3**.
- Q1 (25th percentile) is **71.75**.
- Median (Q2) is **80.5**.
- Q3 (75th percentile) is **89**.
- Maximum value from the review score is **100**.

Table;

Minimum	63.3
Q1 (25th percentile)	71.75
Median (Q2)	80.5
Q3 (75th percentile)	89
Maximum	100

Hours Played:

- Minimum Value of review score attribute is **5**.
- Q1 (25th percentile) is **11**.
- Median (Q2) is **16**.
- Q3 (75th percentile) is **21**.
- Maximum value from the review score is **45**.

Table;

Minimum	5
Q1 (25th percentile)	11
Median (Q2)	16
Q3 (75th percentile)	21
Maximum	45

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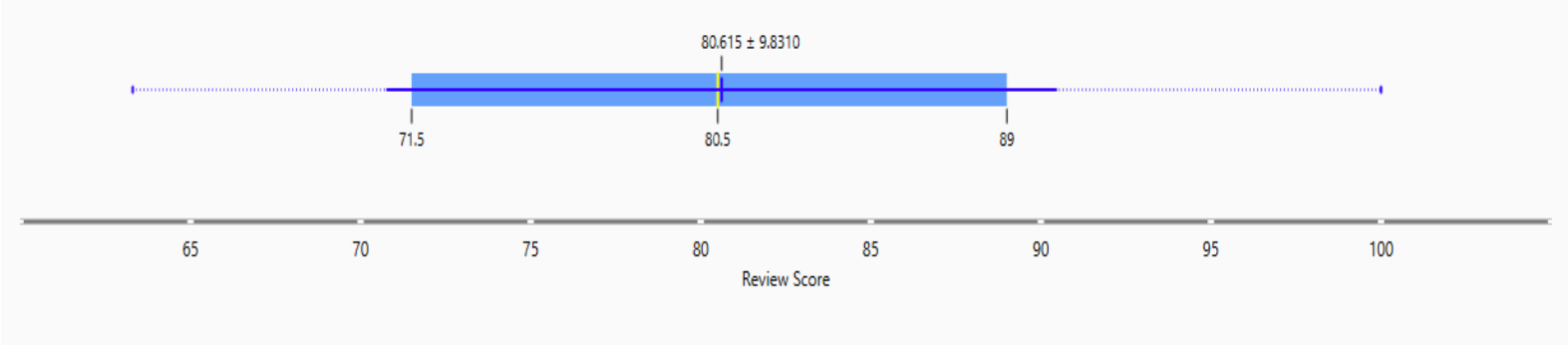
4. Box Plot Visualizations

4.1 Single Box Plot

- The single box plot of **Review Score** shows the distribution of players' ratings for the video game.
- Interpretation:
    1. The **median (Q2)** value is around **80.5**, indicating that most players rated the game positively.
    2. The **interquartile range (IQR = Q3 − Q1)** is approximately **17.5** (from ~71.5 to 89), which suggests **moderate variability** among review scores.
    3. The **whiskers** extend roughly from 71.5 to 100, showing the overall range of scores.
    4. **No significant outliers** are visible, meaning most players gave consistent ratings.
    5. The data distribution appears slightly **left-skewed**, as the lower whisker is slightly longer, indicating a few lower review scores.

- Box Plot Screenshot:



4.2 Grouped Box Plot (if applicable)

- Since the Video Game dataset does not contain a categorical grouping variable (such as genre or region), a grouped box plot is not applicable for this data set.

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5. Correlation Analysis

5.1 Correlation Coefficient Used

- Pearson Correlation is used in this analysis because both attributes **Hours Played** and **Review Score** are Continuous(numeric) variables that show an approximately linear relationship.

5.2 Calculation

- Formula Used:

$$r = \frac{n(\sum xy) - (\sum x)(\sum y)}{\sqrt{[n\sum x^2 - (\sum x)^2][n\sum y^2 - (\sum y)^2]}}$$

x = Hours played
y = Review Score
n = Total no. of observations

- Final correlation value is +0.007.

5.3 Interpretation

The computed correlation coefficient is **+0.007**, which is extremely close to **0**.

- This indicates no linear relationship between **hours played** and **review score.**
- In other words, spending more (or fewer) hours on the game does not significantly affect how highly a player rates it.
- The direction is **slightly positive,** but the value is so small that it is statistically negligible.
- Possible explanation: players may have rated the game based on enjoyment or quality rather than total playtime.

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6. Insights and Conclusion

- **<u>Key insights from Box Plot:</u>**

    1. In the review score attribute's box plot shows that most players gave ratings between **71.5** and **89**, with median **80.5**.
    2. This indicates the majority of review scores are above average and distribution appears slightly **left-skewed**, that means only some players gave lower ratings compared to the majority.
    3. No significant outliers means that feedback was consistent among most players.

- **<u>Key insights from correlation analysis:</u>**
    1. Pearson correlation was used between Hours Played and Review Scores, which comes out to be +0.007.
    2. Here's coefficient value is near to **0** that means no **meaningful linear relationship** between the two variables.
    3. That means how much time is spent by a player does not **significantly influence** how high they rate the game. It means it depends upon game graphics, difficulty level rather than time spent.

- **<u>The reliability of my findings</u>**:

    1. The findings are reliable as the data was analyzed using **Orange Data Mining software**, which applies accurate statistical methods (Pearson correlation and box plots).

    2. However, since the dataset is relatively **small (60 records)** and focused on one specific game, the results might **not generalize** to all gaming scenarios.

    3. The analysis assumes a **linear relationship**, so any **non-linear patterns** between playtime and satisfaction were not captured.

- **<u>Limitations</u>**:

    1. The dataset lacked categorical variables (like game genre or player region), so **group comparisons** could not be made.

    2. The **correlation value was very low**, possibly because **subjective factors** (enjoyment, preference) affect reviews more

than playtime.

3. Outlier impact was minimal, but larger datasets could reveal stronger patterns.

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7. Attachments

- Dataset screenshot (first 5-10 rows)

A	B	C
Player ID	Hours Played	Review Score
Player_1	11	67
Player_2	24	91
Player_3	19	100
Player_4	15	92
Player_5	12	82
Player_6	11	89
Player_7	23	70
Player_8	15	64
Player_9	15	88
Player_10	8	75

End of Report